Speaker Recognition for noisy speech in telephonic channel

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***Abstract* - Speaker recognition in noisy conditions is still a big challenge for researchers. The accuracy of identification rate (IR) is still issue, if the utterance of speech sample is less. This paper aims to implement speaker recognition for telephonic speech samples using MFCC-VQ model and MFCC-LPC with neural network model for text dependent and text independent phrases. The accuracy of MFCC-VQ and MFCC-LPC for telephonic speech sample in normal condition is 57.7% to 63.29% and 60.64% to 68.11%. However, the accuracy through our proposed approach for MFCC-VQ and MFCC-LPC is 75.12% to 88.5% and 84.86 to 90%. We have used NOIZEUS dataset to test speaker recognition on different noisy conditions.**

Keywords – Identification rate, MFCC, VQ, LPC.

I. INTRODUCTION

The uniqueness of physiological traits of human beings made researchers to work on speaker recognition. There are physiological differences between humans, which is used for recognition task. The same phrase spoken by different people will always be different because they will have different pitch loudness and frequency parameters. Even the same phrase spoken by a person at different times will be same physiologically but different mathematically. The working of vocal tracts is the key for speaker verification [1]. The verification of speaker is made on the factor of closeness of two different samples.

The ASV systems are of two types: text dependent and text-independent. Text-independent systems consider the different phrases spoken at the testing and training phases. Text-dependent systems consider the same phrase as input and target speaker. Text dependent speaker verification considers high level features of voice samples such as lexical and phonetic features [1]. The speaker recognition is challenging in telephone network due to noisy environment and short utterance. To identify speaker from Television and telephone samples are difficult due to less availability of speaker's data for training purpose. The background and environment of training and testing data are very different. The noise in telephonic environment always asks questions on accuracy of speaker recognition. The noise can make a significant change in Euclidean distance, while pattern matching. The accuracy of text dependent is generally more due to fix pattern of testing and training data. The work of speaker recognition started in 1960. Filter banks and spectrograms were used in the recognition system for pattern matching [2]. Bricker et al. [3] used auto correlation method for text independent phrases. The use of GMM models for text-independent speaker recognition was investigated by Reynolds et al. [4]. MFCC and segmental approach was studied in [5]. The work on noisy environment has not been explored significantly. Factor Analysis (FA) [6] and Universal Background Model (UBM) [7] are statistical approaches used for speaker recognition. Support Vector machine (SVM) [8] and Joint factor analysis (JFA) [9] are used for short utterances. We used NOIZEUS dataset to perform our result in noisy environment.

The performance and robustness of experiments are evaluated in terms of, Identification rate (IR), Accuracy, and Error rate. The previous work on speaker recognition is done mostly on normal environment. This work aims to compute results for tough conditions where utterance (speech data) can be shorter and environment possibility be noisy. The waveform of noisy speech for street, car and telephone is shown in Fig.1, 2 and 3.

The Section II in the paper briefly describes the fundamental component of work and algorithmic steps of proposed framework. Results are given in section III following concluding remarks, given in section IV.

II. FUNDAMENTAL COMPONENTS

This part of paper contains the description of algorithm used in proposed approach. It also reveals their usefulness and robustness towards human's speech identification system. The algorithms used in our work are MFCC [10], VQ [11], LPC [12], Wiener filter [13], artificial neural network [14].

A. *MFCC*

The speech signal is first digitized and then converted into frames. The length of frames can be in range between 20

to 40 ms. After framing the speech signal is smoothed by Hamming window. MFCC involves the following five steps:

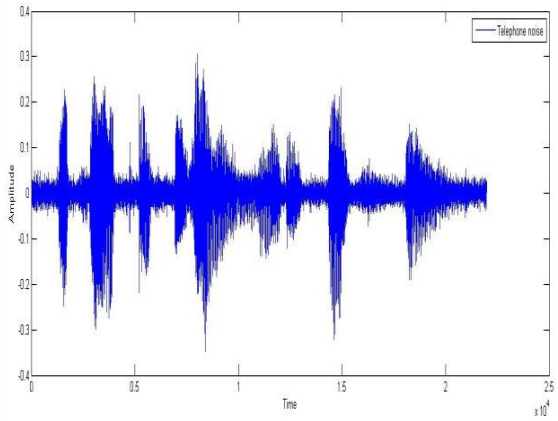


Fig.1

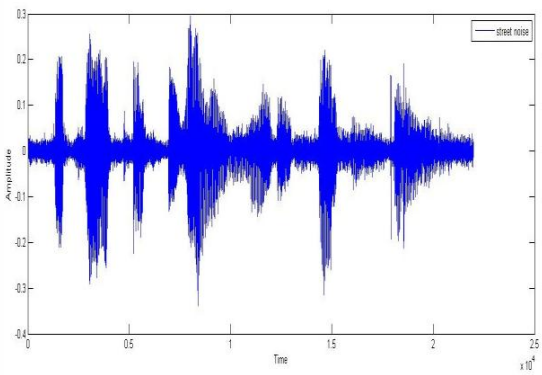


Fig.2

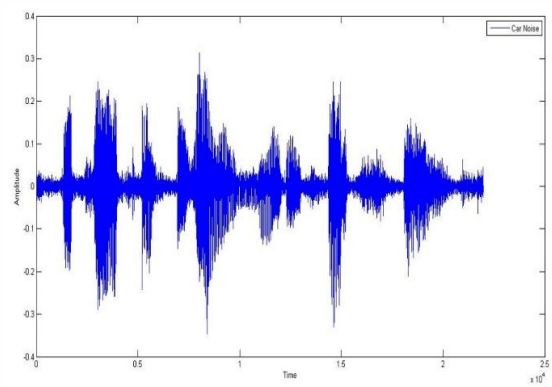


Fig.3

1) The Fast Fourier transformation of speech signal x(n) is done using a 30 ms Hamming window H(n). Speech signal and hamming window is used to calculate power spectral estimation. The power spectral estimation is given by eq (1).

$$\dot{x}(k) = \sum_{n=0}^{n-1} x(n)H(n)e^{-j2\pi nk/N} \qquad 0 \le n \le N \qquad (1)$$

where N is hamming window of length 30 ms and having cosine impulse given by eq (2).

$$H(n) = \begin{cases} 0.54 - 0.46\cos\left(\frac{2\pi n}{N-1}\right) & for \;\; n = 0, \dots N-1 \\ 0 & else \end{cases} \qquad (2)$$

To show the characteristics of vocal tract, square magnitude of DFT is calculated by eq (3).

$$X_k = |\dot{x}(k)|^2 \qquad (3)$$

2) Apply triangular filters to power spectrum, energy is calculated in each filter bank channel by eq (4)

$$E_m = \sum_{k=0}^{k-1} \emptyset_m(k) X_k; \quad m = 1, 2, \dots M \qquad (4)$$

Where M is number of triangular filters (∅), which can have range 20 to 35 with condition given by eq (5)

$$\sum_{k=0}^{k-1} \emptyset_m(k) = 1 \qquad \forall m \qquad (5)$$

3) Cepestral coefficient is calculated by computing discrete cosine transformation of log filterbank energies given by eq (6)

$$c_j = \sum_{m=1}^{M} \log_{10}(E_m).\cos j((m+0.5)\frac{\pi}{m}) \quad j = 1, 2 \dots L \; (6)$$

The weighted value can be calculated by eq (7)

$$c_i = (1 + \frac{l}{2}\sin(\frac{\pi i}{l}))c_i \qquad (7)$$

where c_i is cepestral coefficient of ith order and liftering coefficient (l) is set to 21 in our approach.

4) 13-Dimensional feature vector is produced after normalized frame energy.

5) First and second order derivative of 13 coefficient are calculated. These are append to original MFCC, 39 dimensional feature vectors will be produced for each frame.

B. *VQ*

VQ is distribution of prototype vectors which models probability density function of speech signals. The same number of points is kept in groups, which are made by

dividing large set of point works by dividing a large set of points. Larger and high-dimensional data are matched according to density of data. The centroid of data helps to represent data points according to their index.

VQ suits for lossy data compression. A vector quantizer maps d-dimensional vectors in the vector space R_k into a finite set of vectors $Q = \{q_i : i = 1, 2,..., N\}$. Each vector q_i is called a codeword or a codevector and the set of all the code vectors are called a codebook. The region of nearest neighbor q_i is called Voronoi region.

C. *LPC-MFCC*

LPC is one of the effective techniques for feature extraction. The vocal tract information can be extracted by LPC. The steps involve in LPC are shown in fig.

The steps are

1) Power Spectrum of speech is followed by frame blocking and windowing.
2) Set sampling rate 16 kHz by grouping all critical bands in bark scale.
3) Cube root compression and loudness equalization is calculated for power law of hearing
4) Perform Inverse fast Fourier transformation.
5) LPC analysis is made by -Durbin procedure.
6) LPC coefficients are converted into cepstral coefficients.

The frequency in bark and frequency in Hz are related in eq (8).

$$f(bark) = 6 * \text{arcsinh}\left(\frac{f(Hz)}{600}\right) \quad (8)$$

Algorithm (MFCC-VQ)

Initialization: s is signal to be analyzed.

fs is sampling rate (12500 Hz)

m is distance between two frames (100)

n is number of samples each frames (256)

p is number of filters in filterbank

Step 1: Record the audio using microphone and store it into train folder for training purpose.

Step 2: Record the audio for testing using microphone and save it into test folder.

Step 3: Put the signal into frames (s, fs, m, s)

Step 4: Call wiener filter to remove noise (n, fs, s)

Step 5: Feature Extraction is done using MFCC

MFCC (n, fs, s)

Step 6: Compute matrix for mel- space filterbank.

Melfb (p, n, fs)

Compute MFCC of audio data for test and train samples

MFCC (train{i})

MFCC (test(i))

Step 7: Vector quantization model is used for classifying samples from MFCC.

7.1 Vqlbg (MFCC_train {i})

7.2 Vqlbg (MFCC_test {i})

Step 8: Compute Euclidean distastance between (7.1 and 7.2)

(v, vqlb_test(i,i))

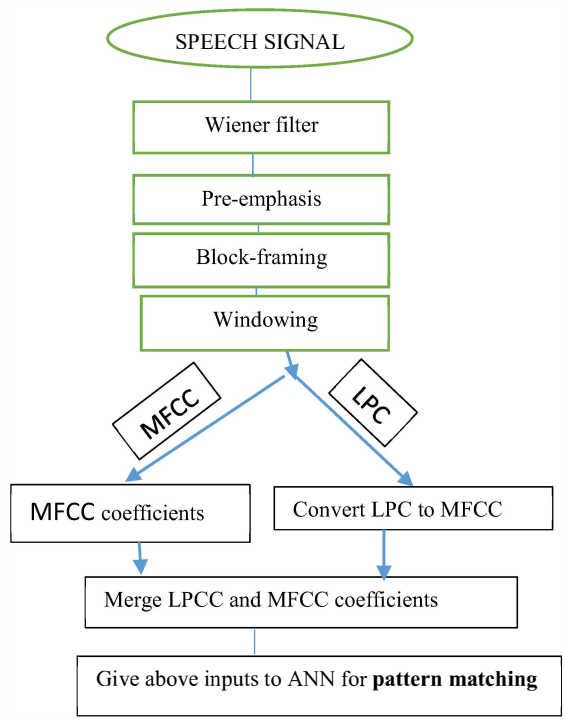
Step 9: Pattern matching

Min(Euclidean distance (vqlbg_test(i))) && min (distance)< threshold

Result = recognized, else

Result= not recognized.

Flowchart of MFCC-LPC



III. RESULTS

The work is performed on 15 no. of speaker's dataset. There are total 10 male and 5 female speakers. The data is taken on real time basics. The table 1 gives recognition result on normal environment. The table 2 is result computed on noisy telephonic environment without using filters. The table 3 gives the result by our approach using wiener filter for telephonic speech samples. The recognition is done using MFCC-VQ and MFCC-LPC with neural network. The result is computed on text dependent and text independent phrases.

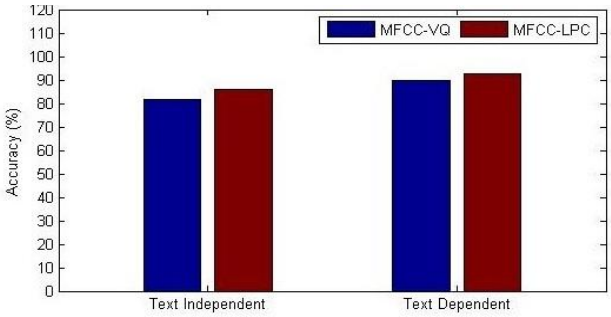


Fig. 4

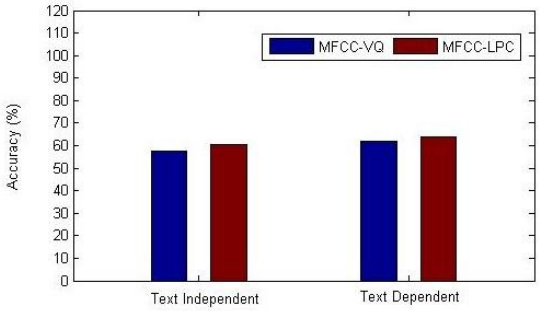


Fig. 5

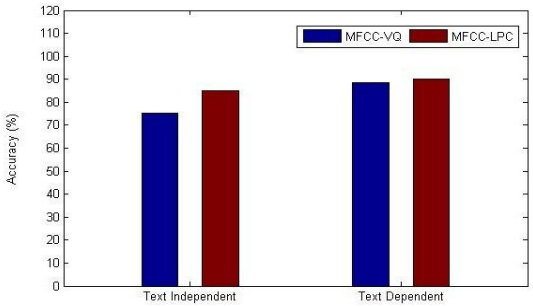


Fig. 6

Table I.

Speakers		M1	M2	M3	M4	M5	M6	M7	M8	M9	M10	F1	F2	F3	F4	F5	Average
Text independent	MFCC-VQ	80	76.67	83.3	70	70	73.33	70	86.67	90	66.67	80	76.67	70	73.33	86.67	81.64
	MFCC-LPC	83.33	90	90	86.67	73.3	86.67	83.33	86.67	90	86.67	83.3	83.3	90	86.67	90	86
Text dependent	MFCC-VQ	90	93.3	90	100	83.3	83.3	86.7	100	100	83.3	90	83.3	90	90	83.3	90
	MFCC-LPC	100	100	90	93.3	90	90	100	86.67	90	90	93.3	100	86.7	86.7	90	92.5

Table II.

Speakers		M1	M2	M3	M4	M5	M6	M7	M8	M9	M10	F1	F2	F3	F4	F5	Average
Text independent	MFCC-VQ	66.6	63.33	66.66	63.33	60	63.33	60	50	50	50	50	63.33	60	50	50	57.76
	MFCC-LPC	70	70	63.33	50	50	50	63.33	66.6	66.6	63.3	63.3	50	50	66.6	66.6	60.64
Text dependent	MFCC-VQ	66.6	66.6	66.6	70	70	50	70	66.6	66.6	50	50	66.6	63.3	63.3	63.3	63.29
	MFCC-LPC	70	70	66.6	66.6	66.6	70	66.6	50	66.6	83.3	70	83.3	66.6	63.3	63.3	68.11

Table III.

Speakers		M1	M2	M3	M4	M5	M6	M7	M8	M9	M10	F1	F2	F3	F4	F5	Average
Text independent	MFCC-VQ	70	76.67	83.3	66.67	66.67	73.33	70	86.67	83.33	66.67	80	76.67	70	73.33	83.3	75.12
	MFCC-LPC	83.33	83.3	83.3	86.67	73.3	86.67	83.33	86.67	90	83.3	83.3	83.3	90	90	86.67	84.86
Text dependent	MFCC-VQ	93.3	83.7	90	90	83.3	83.3	86.7	93.3	100	83.7	90	83.3	83.7	90	83.3	88.5
	MFCC-LPC	90	90	93.3	93.3	90	86.67	90	86.67	90	93.3	93.3	86.67	86.67	90	90	90

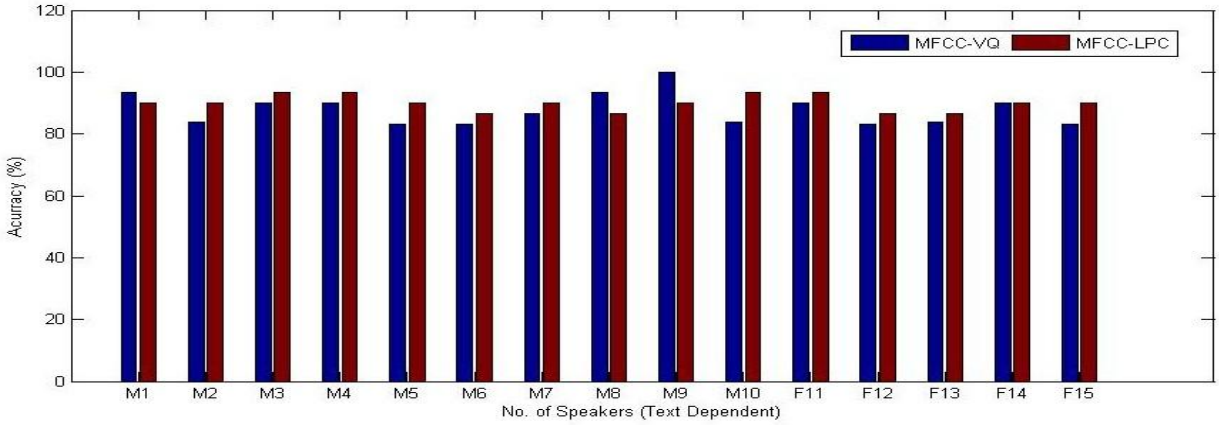


Fig. 7

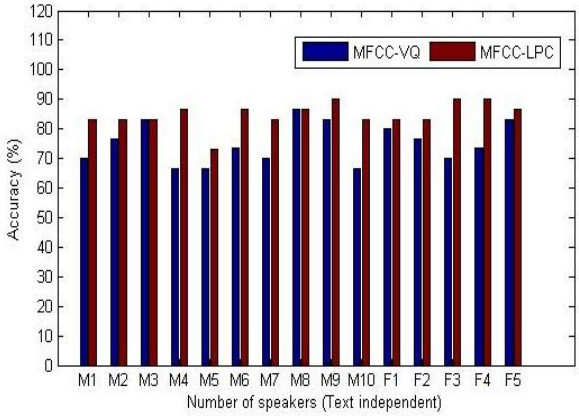


Fig. 8

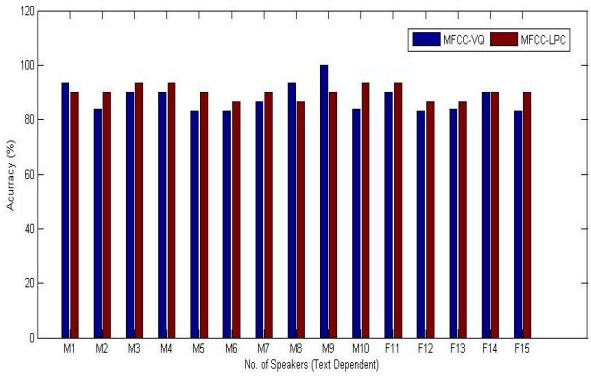


Fig. 9

The accuracy of speaker recognition in normal condition (Fig 4) is 81.64% and 90% for text dependent recognition using MFCC-VQ and 90% to 92% using MFCC-LPC fusion. However, in noisy (12dB) condition using both methods the recognition rate decreases remarkably to 23% to 25% (Fig 5).

The recognition rate for telephonic speech is 79.12% to 88.5% and 84.86% to 90% using MFCC-VQ and MFCC-LPC by our approach (fig. 6). The identification rate of each speaker is shown in Fig. 7, 8, 9. The accuracy is calculated by the following eq (9).

$$Accuracy = \frac{no.of\ correct\ recognition}{total\ no.of\ trails} \times 100 \quad (9)$$

IV. CONCLUSION

The accuracy of speaker recognition in noisy environment is still a big issue for researchers. The noise removal filters are useful to cut unwanted sounds affecting recognition rate. We have seen with the use of wiener filter, the recognition rate has increased remarkably. However, there is always scope of improvement in accuracy of speaker recognition. The recognition is tough when utterance is shorter in length.

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